



Practice Graded Assignment: Course project

Review fellow learners

Congrats on submitting your assignment! You have reviewed enough peers, but you can help even more learners complete this course by continuing to give reviews.

[Review assignments](#)

🌟 AI Grading

Your assignment has been graded by AI. View your scores below. You'll also find feedback from Coursera AI, an AI-powered learning assistant, which provides insights on how to enhance your work.

Instructions My submission Peers to review Discussions

In this project, you will demonstrate your ability to perform the following:

- Create custom analytics rules to detect threats and anomalous behaviors in your environment.
- Configure and manage Azure Monitor to monitor the secure operation of Azure solutions.
- Enable alerts for proactive response to critical conditions.

You are required to create two deliverables that will detail the steps and screenshots of the activities related to detecting vulnerabilities and notifying about unavailable virtual machines (VMs). You can create these deliverables using Microsoft Word or a similar application.

These deliverables are:

- *Log4j vulnerability detection project solution*
- *VM unavailability notification project solution*

Your detailed tasks will include the following:

Task 1/2: Create a rule to detect Log4j vulnerability from a template

- Install the *Log4j Vulnerability Detection* solution from the Content hub.
- Navigate to the *Log4j vulnerability exploit aka Log4Shell IP IOC* scheduled analytics rule from the Rule templates tab.
- Create a rule logic for *Log4j vulnerability exploit aka Log4Shell IP IOC*, and configure settings.

Task 2/2: Create an alert rule to notify VM unavailability

- Set the scope in the metrics explorer for a subscription or resource group with the VM to monitor.
- Create an alert rule and action group.

At the end of this project, you will have your version of the following deliverables for submission:

- *Log4j vulnerability detection project solution.docx*
- *VM unavailability notification project solution.docx*

As part of this project, you will also review the project submissions of two peers using the peer review rubric provided.

You should plan to complete this course project in about thirty minutes. Also, allow up to one hour to complete the graded quiz that follows this project.

Refreshing your knowledge

Before starting on the project or while doing the project, you can refresh your knowledge by referring to the videos, readings, and associated exercise/solution documents in the below lessons:

- Week 1, Lesson 2: Explore Azure Monitor
- Week 1, Lesson 4: Collect virtual machine event and performance data using Azure Monitor
- Week 3, Lesson 3: Investigate threats

Project scenario

In this project, you will go through the typical activities that an Azure security engineer would perform in an organization.

Imagine you've recently joined a large organization as an Azure security engineer. You are responsible for acting on two problem statements:

Problem statement 1: Your company has several Java applications running on production, and the recent Log4j vulnerability has caused much concern. Stakeholders are worried about the safety of these applications in the environment.

As a security engineer, you're tasked with setting up a Microsoft Sentinel analytics rule from a template to proactively detect Apache Log4j vulnerability exploits across your environment. You'll need to create rules to frame user accounts and IP addresses in your logs as trackable entities. These rules will identify notable information in the alerts generated by the rules and package them as incidents to be investigated.

Task 1: Create a rule to detect Log4j vulnerability from a template.

Problem statement 2: A mission-critical application with financially sensitive operations that must be completed for the upcoming IPO release is running on a VM. The stakeholders need to be alerted when the VM is not functional.

You are tasked with monitoring the VM and creating an alert if it becomes non-functional. You will have to create an alert rule in Azure Monitor using the VM availability metric to notify you when the VM is unavailable. You will also need to create an action group to proactively receive SMS notifications whenever an alert is generated.

Task 2: Create an alert rule to notify VM unavailability.

Setting up the system

This project consists of tasks you should complete on the Azure platform. Before you begin, you must have the following:

- Access to Azure Monitor and Microsoft Sentinel to detect a vulnerability and notify VM unavailability
- An application, such as Snipping Tool, to capture screenshots
- An application, such as Microsoft Word, to create the two deliverables: *Log4j vulnerability detection project solution* and *VM unavailability notification project solution.docx*

Guidelines for the deliverables

We've provided two templates to help you document your work and create your deliverables. The templates provide starting points for the tasks you need to complete.

1. Select the template titled *Log4j vulnerability detection project solution template*.



Log4j vulnerability detection project solution template
DOCX File

2. Rename the downloaded template with a unique file name so you can easily find it and upload it for submission later. For example, "*Log4j vulnerability detection project solution_Your initials*."

3. Select the template titled *VM unavailability notification project solution template*.



VM unavailability notification project solution template
DOCX File

4. Rename the downloaded template with a unique file name so you can easily find it and upload it for submission later. For example, "*VM unavailability notification project solution_Your initials*."

Guidelines for screenshots

1. Use an application, such as Snipping Tool, to capture screenshots.
2. Perform a step in the task and capture the corresponding screenshot.
3. Confirm that the screenshot displays the result of your activity. You should set your system's display resolution to the highest resolution available.
4. You may also want to mask or remove personal information, such as sign in credentials, from the screenshot.
5. Copy and paste the screenshot below the appropriate step in your version of the relevant deliverable: *Log4j vulnerability detection project solution* or *VM unavailability notification project solution*.

^ Completing project tasks (1/2): Create a rule to detect Log4j vulnerability from a template

Creating a well-formatted document with steps and screenshots will help your peer effectively review your submission. Follow the instructions below to create your project deliverable.

Deliverable: Log4j vulnerability detection project solution

Task: Create a rule to detect Log4j vulnerability from a template

Activity 1: Install the *Log4j Vulnerability Detection* solution from the Content hub.

While performing this activity, document each step and the associated screenshot in your version of the *Log4j vulnerability detection project solution* document.

Remember to include the following:

- A step for selecting the **Content hub** option.
- A step for selecting the **Log4j Vulnerability Detection** solution.
- A screenshot that shows the **Install** option on the **Log4j Vulnerability Detection** wizard.

Activity 2: Navigate to the *Log4j vulnerability exploit aka Log4Shell IP IOC* scheduled analytics rule from the Rule templates tab.

While performing this activity, document each step and the associated screenshot in your version of the *Log4j vulnerability detection project solution* document.

Remember to include the following:

- A step or screenshot that indicates selecting the **Rule templates** tab.
- A screenshot that indicates selecting **Log4j vulnerability exploit aka Log4Shell IP IOC**.

Activity 3: Create a rule logic for *Log4j vulnerability exploit aka Log4Shell IP IOC*, and configure settings.

You can use the following information while performing this activity:

Note: It is not mandatory that you follow the specific names or settings. However, following the details provided will make it easier when you conduct peer reviews or compare your work with the exemplar you'll download at the end of this project.

Parameter	Value
Key field	Timestamp
Value	Timestamp
Alert Name Format	Log4j vulnerability exploit aka Log4Shell IP IOC from {{timestamp}}
Automation rule name	Log4J vulnerability exploit detection
-	-

lag	Log4J exploit
-----	---------------

While performing this activity, document each step and the associated screenshot in your version of the *Log4j vulnerability detection project solution* document.

Remember to include the following:

- A step or a screenshot that indicates selecting the **Create rule** option on the **Log4j vulnerability exploit aka Log4Shell IP IOC** wizard.
- A step or screenshot that indicates **Log4j vulnerability exploit aka Log4Shell IP IOC from {{timestamp}}** entered in the **Alert Name Format** box.
- A screenshot that indicates **Log4J vulnerability exploit detection** entered in the **Automation rule name box** in the **Create new automation rule** wizard.
- A screenshot that shows the **Review and create** tab on the **Analytics rule wizard – Create a new Scheduled rule** page with the **Save** option.

^ Completing project tasks (2/2): Create an alert rule to notify VM unavailability

Creating a well-formatted document with steps and screenshots will help your peer effectively review your submission. Follow the instructions below to create your project deliverable.

Deliverable: VM unavailability notification project solution

Task: Create an alert rule to notify VM unavailability.

Activity 1: Set the scope in metrics explorer for a subscription or resource group with the VM to monitor.

While performing this activity, document each step and the associated screenshot in your version of the *VM unavailability notification project solution* document.

Remember to include the following steps/screenshots:

- A step or a screenshot that indicates selecting **Metrics**.
- A screenshot that shows **Virtual machines** selected as the **Resource type** in the **Select a scope** wizard.

Activity 2: Create an alert rule and action group.

Problem statement recap: You will have to create an alert rule in Azure Monitor using the VM availability metric. You will also need to create an action group to proactively receive SMS notifications whenever an alert is generated.

You can use the following information while performing this activity:

Note: It is not mandatory that you follow the specific names or settings. However, following the details provided will make it easier when you conduct peer reviews or compare your work with the exemplar you can download at the end of this project.

Parameter	Value
Action group name	My Action Group 1
Display name	My Group
Name	Text administrator
Severity	3 - Informational
Alert rule name	VM unavailability

While performing this activity, document each step and the associated screenshot in your version of the *VM unavailability notification project solution* document.

Remember to include the following screenshots:

- A screenshot that shows the desired **Metric** selected.
- A screenshot that shows values selected in the **When to evaluate** section of the **Create an alert rule** page.
- A screenshot that shows the desired notification type selected.
- A screenshot that shows the alert rule created on the **Alert rules** page.
- A screenshot that shows the action group created on the **Action groups** page.

^ Finalizing your assignment

After completing all the tasks, you can finalize your deliverables. Ensure that you've added the steps and screenshots for all the tasks.

Remove any instruction sections or examples provided in the original deliverable templates. Your final deliverables should contain only your work. You are then ready to upload your deliverables in the **My submission** tab.

^ Review criteria

Fifteen points are allotted for this final course project. The points are distributed across the various tasks and activities that you have completed while developing your version of the following deliverables:

- *Log4j vulnerability detection project solution* (9 pts)
- *VM unavailability notification project solution* (6 pts)

Your grade will be based on the following distribution of points across tasks and activities.

Deliverable 1: Log4j vulnerability detection project solution

Task 1/2: Create a rule to detect Log4j vulnerability from a template. (9 pts)

Activity 1: Install the *Log4j Vulnerability Detection solution* from the Content hub. (3 pts)

Activity 2: Navigate to the *Log4j vulnerability exploit aka Log4Shell IP IOC* scheduled analytics rule from the Rule templates tab. (2 pts)

Activity 3: Create a rule logic for *Log4j vulnerability exploit aka Log4Shell IP IOC*, and configure settings. (4 pts)

Deliverable 2: VM unavailability notification project solution

Task 2/2: Create an alert rule to notify VM unavailability. (6 pts)

Activity 1: Set the scope in metrics explorer for a subscription or resource group with the VM to monitor. (2 pts)

Activity 2: Create an alert rule and action group. (4 pts)

^ Preparing for project submission

Note: Do not proceed to the **My submission** and **Peers to review** tabs until you've completed your version of the following deliverables:

- *Log4j vulnerability detection project solution*
- *VM unavailability notification project solution*

Review, proofread, and edit your deliverables. Ensure they are ready to submit for peer review.

To submit the deliverables, upload your documents to the Coursera platform following the prompts in the **My submission** and **Peers to review** tabs.

Peer review

Giving, receiving, and incorporating feedback is a crucial professional skill. After you submit your deliverable, as the next activity, you will have to review and provide feedback for two peers' submissions. Similarly, your submission will be reviewed by a peer. We've shared a rubric to ease the review process. You will be presented with questions about your peers' submissions. For example, did your peer document all the key evidence? If prompted, you may have to provide more in-depth, written feedback for some questions. Please provide constructive feedback to help your peers understand what they did well and what they can improve on.

Compare your work (Discussion)

After submitting your deliverables and completing the two peer reviews, you will be able to access the following exemplars:

- *Log4j vulnerability detection project solution exemplar*
- *VM unavailability notification project solution exemplar*

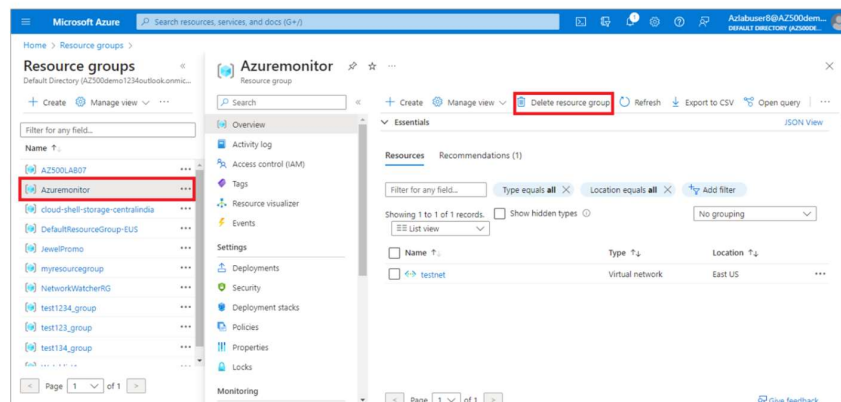
These exemplars are step-by-step guides for the project tasks. You can compare the exemplars with your submissions. You can then choose to reflect on your learning and share your assignment experience with peers in the discussion forum.

After completion of the project

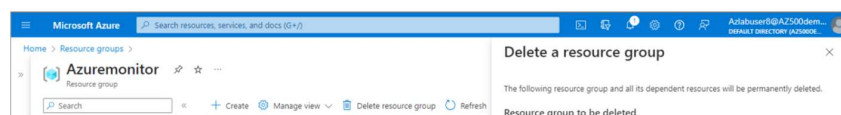
When working on your Azure subscription, at the end of a project, it's good practice to assess whether you still require the resources you created. Running resources can cost you money. You can delete resources individually or delete the entire resource group containing the resources. Here's how you can do this.

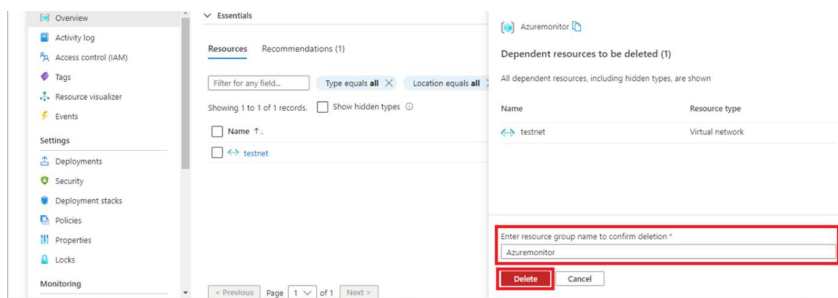
Clean up resources

1. In the Azure portal, navigate to **Resource groups** to view the resource group you created for this exercise.
2. Select the resource group.
3. To delete the entire resource group, select **Delete resource group**.



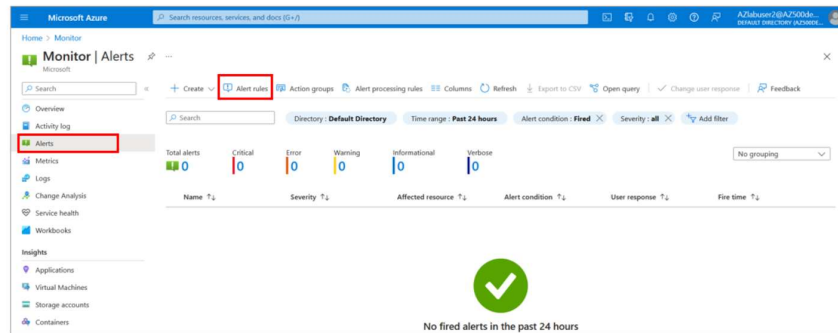
4. In the **Delete a resource group** wizard that appears, enter the resource group name to confirm deletion. Then, select **Delete**.



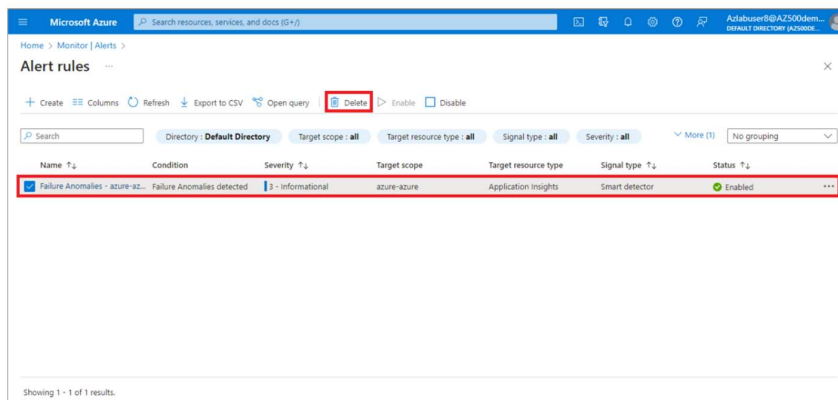


Delete the alert rule

1. In the Azure portal, in the global search bar, search for and select **Monitor**.
2. From the **Monitor** menu, select **Alerts**.
3. On the **Alerts** page, select **Alert rules**.

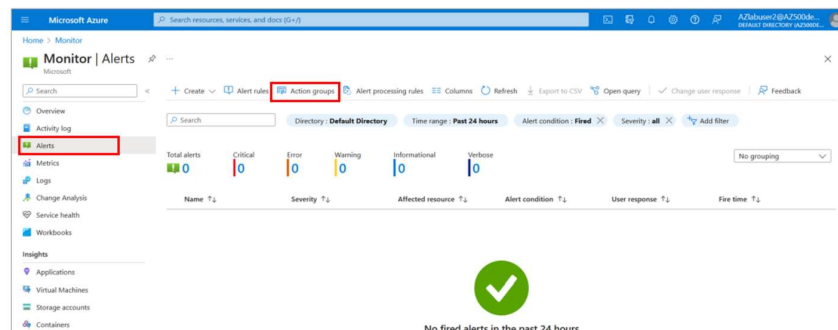


4. Select the created alert rule and then select **Delete**.

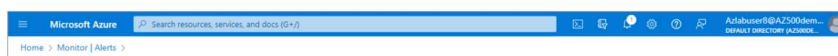


Delete the action group

1. In the Azure portal, in the global search bar, search for and select **Monitor**.
2. From the **Monitor** menu, select **Alerts**.
3. On the **Alerts** page, select **Action groups**.



4. Select the created action group and then select **Delete**.



Action groups

Create

Columns

Refresh

Open query

Delete

Enable

Disable

Test action group

Search

Directory: Default Directory

Resource group: all

Location: all

Status: Enabled

No grouping

Name	Short name	Resource group	Subscription	Actions	Status	
My Action Group 1	My Group	myresourcgroup	Azure Pass - Sponsorship	1 SMS message	Enabled	...

Showing 1 - 2 of 2 results.

Like

Dislike

Report an issue

Go to next item →